Category

Forward Commitment and Contingent Claim Features and Instruments

Related Questions:

Practice questions related to this topic

OPTIONS

Learning Outcomes

- define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics
- determine the value at expiration and profit from a long or a short position in a call or put option
- contrast forward commitments with contingent claims

Contingent claims are a type of ETD or OTC derivative contract in which one of the counterparties has the right to determine whether a trade will settle based on the underlying value. Option contracts are the most common contingent claim. Similar to forwards and futures, options are derivative contracts between a buyer and a seller that specify an underlying, contract size, a pre-agreed execution price, and a maturity date. The option buyer has the right but not the obligation to transact the trade, and the option seller has the obligation to fulfill the transaction as chosen by the option buyer. As a consequence, the payoff to an option buyer is always zero or positive. It can never be negative.

Assume an option buyer pays a premium (c_0) of \$5 at $t = 0$ for the right—but not the obligation—to buy stock S at time T at a pre-agreed price (X) of \$30. The option buyer's decision at maturity depends on the stock price at maturity (S_T), as shown in the following two scenarios:

Scenario 1: Transact ($S_T > X$)

- If S_T = \$40, the option buyer chooses to exercise the option and buy the stock for X = \$30.
- The option buyer gains \$10 [($S_T - X$) = \$40 – \$30] on the transaction.
- The option buyer realizes a \$5 profit [($S_T - X$) – c_0 = (\$40 – \$30) – \$5].

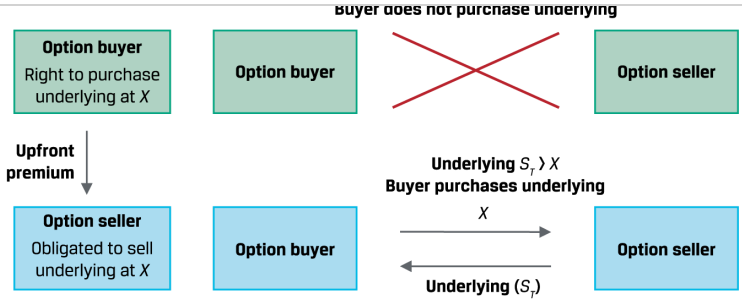
Scenario 2: Do Not Transact ($S_T < X$)

- If S_T = \$25, the option buyer chooses not to exercise the option and buy the stock for X = \$30.
- The option buyer realizes a \$5 loss (c_0 = \$5).

Exhibit 6 shows the option mechanics for the case where the option buyer pays for the right to purchase an underlying stock S at a pre-agreed execution price of X in the future ($t = T$).

Exhibit 6:

Call Option Mechanics



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The decision to transact the underlying is referred to as an **exercise**, and the pre-agreed execution price is called the **exercise price** (or strike price). Option buyers may transact the underlying in the future at their sole discretion at the exercise price, a pre-agreed future spot price that may be above, at, or below the forward price, as shown in an earlier lesson. This right to exercise in the future has a value that is paid upfront to the option seller in the form of an **option premium**.

Option contract terms, such as the right to buy or sell, exercise price, maturity, and size, may either be agreed on between the counterparties in an over-the-counter transaction or executed on an exchange based on standardized terms. This lesson focuses on **European options**, or options that may be exercised only at maturity, although other option styles exist. **American options**, for example, may be exercised at any time from contract inception until maturity. Note the labels "European" and "American" refer not to where these options are used but, rather, to the difference in when they can be exercised.

Two primary option types exist—namely, (1) the right to buy an underlying known as a **call option** and (2) the right to sell the underlying, or a **put option**. An option buyer will exercise a call or put option only if it returns a positive payoff. If not exercised, the option expires worthless, and the option buyer's loss equals the premium paid.

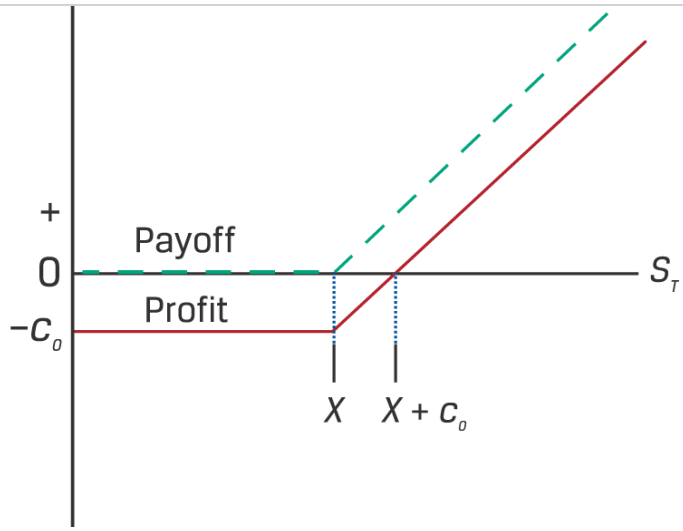
One factor in an option's value prior to maturity ($t < T$) is the option's exercise value at time t, which is referred to as an option's **intrinsic value**.

We can say a call option is **in-the-money** at time t if the spot price, S_t, exceeds X, with an intrinsic value equal to ($S_t - X$). Both **out-of-the-money** options (where $S_t < X$) and **at-the-money** options ($S_t = X$) have zero intrinsic value, so their price, c_t, consists solely of time value.

Call option buyers will gain from a rise in the price of the underlying. Exhibit 7 shows the payoff and profit at maturity for a call option buyer.

Exhibit 7:

Long Call Payoff Profile



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The option buyer pays a call option premium, c_0, at time $t = 0$ to the option seller and has the right to purchase the underlying, S_T, at an exercise price of X at time $t = T$. The exercise *payoff* ($S_T - X$) is positive if $S_T > X$ and zero if $S_T \leq X$. The call option value at maturity, c_T, may be expressed as follows:

$$c_T = \max(0, S_T - X). \quad (1)$$

The call option buyer's *profit* equals the *payoff* minus the call premium, c_0 (ignoring the time value of money in this lesson):

$$\Pi = \max(0, S_T - X) - c_0. \quad (2)$$

This asymmetric payoff profile is a common feature of contingent claims, which are sometimes referred to as **non-linear derivatives**.

EXAMPLE 5

Hightest Capital purchases a call option on the S&P 500 Health Care Select Sector Index (SIXV). This six-month exchange-traded option contract has a size of 100 index units and an exercise price of $1,240 per unit versus the initial SIXV spot price of $1,180.95. The option premium paid upfront is $24.85 per unit, or $2,485 (= $24.85 × 100). As the option nears maturity, a Hightest analyst is asked to determine the expected option payoff and profit per unit at maturity under different scenarios for the SIXV spot price on the exercise date.

She compiles the following table:

c_0 = Call option premium = $24.85 per unit.

= Exercise price at time T = $1,240 per unit.

S_T = Spot price per unit at time T.

SIXV Spot Price (S_T)	Exercise Price (X)	Payoff $\max(0, S_T - X)$	Profit $\max(0, S_T - X) - c_0$
$1,280	$1,240	$40	$15.15
$1,260	$1,240	$20	−$4.85
$1,240	$1,240	$0	−$24.85
$1,220	$1,240	$0	−$24.85

Value of an option is always positive and declines to zero as an option reaches maturity.

In contrast to the call option buyer with unlimited upside potential and a loss limited to the premium paid, the call option seller receives a maximum of the premium and faces unlimited downside risk as the underlying appreciates above the exercise price. The short call payoff profile in Exhibit 8 is a mirror image of Exhibit 7.

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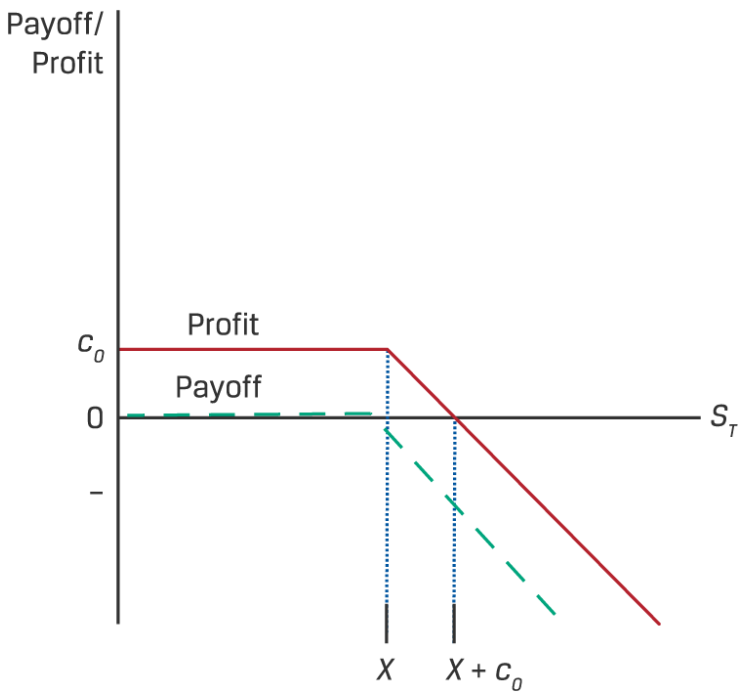
Exhibit 8:

Short Call Payoff Profile



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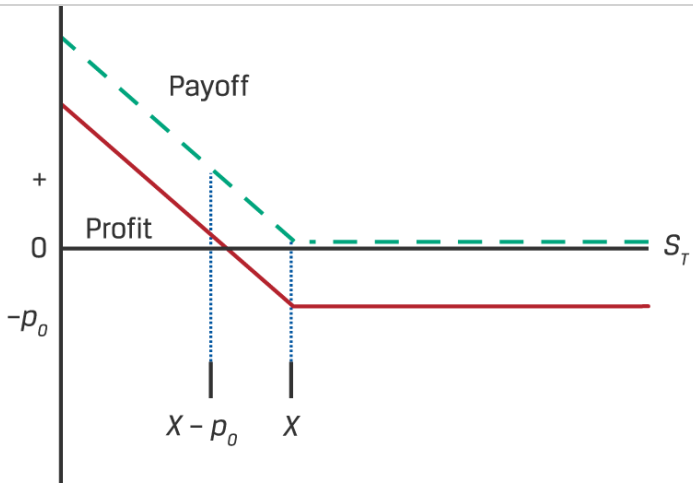
The option buyer and seller payoff profiles demonstrate the one-sided nature of counterparty credit risk for contingent claims. That is, the option seller has no credit exposure to the option buyer once the premium is paid. However, the option buyer faces the counterparty credit risk of the option seller equal to the option payoff at maturity.

Put option buyers benefit from a lower underlying price by selling the underlying at a pre-agreed exercise price. A put option buyer exercises when the underlying price is below the exercise price ($S_T < X$), as shown in Exhibit 9.

Exhibit 9:

Long Put Payoff Profile



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The put option buyer pays a premium of p_0 at inception to the option seller and will exercise the option only if $(X - S_T) > 0$. As in Equations 1 and 2, we may show the long put option *payoff* and *profit*, Π, as follows:

$$p_T = \max(0, X - S_T). \quad (3)$$

$$\Pi = \max(0, X - S_T) - p_0. \quad (4)$$

The payoff or profit from a put option seller's perspective is the opposite of the put option buyer's gain or loss for a given underlying price at expiration. As in the case of the call option seller, the put option seller has a maximum gain equal to the premium. However, although a call option seller faces unlimited potential loss as the underlying appreciates beyond the exercise price, the put option seller's loss is usually limited because the underlying price cannot fall below zero. The short put option *payoff* and *profit* are as follows:

$$-p_T = -\max(0, X - S_T). \quad (5)$$

$$\Pi = -\max(0, X - S_T) + p_0. \quad (6)$$

Question Set: Options

1. Calculate the SIXV spot price at maturity from Example 5 at which Hightest Capital will reach a breakeven point and earn zero profit.

   Hide Solution

   Solution:

   The call purchaser has a profit equal to $\max(0, S_T - X) - c_0$. In the case of Hightest in Example 5, the SIXV exercise price is $1,240 and the initial call premium is $24.85. The breakeven or zero profit point is therefore equal to $1,240 + $24.85, or $1,264.85.

2. A put option seller receives a $5 premium for a put option sold on an underlying with an exercise price of $30. What is the option seller's maximum profit under the contract? What is the maximum loss under the contract?

   Hide Solution

   Solution:

   A put option seller receives a $5 premium ($p_0$) for a put option sold on an underlying with an exercise price (X) of $30. The put option seller's profit is Π =

contract is $5 and the maximum loss under the contract is $25.

3. Identify the option contract participants that correspond to the following statements:

A. Put option seller	1. Has no counterparty credit risk to the option buyer once the upfront premium has been paid
B. Call option seller	2. Earns a profit equal to the premium if the underlying price at maturity is less than the exercise price
C. Both a put option seller and a call option seller	3. Earns a profit equal to the premium if the underlying price at maturity exceeds the exercise price

Hide Solution

Solution:

1. The correct answer is C. An option seller has no counterparty credit risk to the option buyer once the upfront premium has been paid.

2. The correct answer is B. A call option seller earns a profit equal to the premium if the underlying price at maturity is less than the exercise price.

3. The correct answer is A. A put option seller earns a profit equal to the premium if the underlying price at maturity is greater than the exercise price.

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NA Scenario 2 : If ST = $25, the option buyer chooses not to exercise the option and buy the stock for X = $25 not $30. right?

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